Lagrange multiplier $\boldsymbol{\lambda}_{n+1/4}$ (and likewise $\boldsymbol{\lambda}_{n+1}$) can be obtained by solving the linear system

$$\boldsymbol{g}'(\boldsymbol{q}_n)\boldsymbol{M}^{-1}\left[\left(1-\frac{h\gamma}{4}\right)\boldsymbol{p}_n+\sqrt{h\gamma k_{\mathrm{B}}\mathrm{T}\boldsymbol{M}}\boldsymbol{R}_n-\boldsymbol{g}'(\boldsymbol{q}_n)^T\boldsymbol{\lambda}_{n+1/4}\right]=\mathbf{0},$$

which is easily computed once the $\boldsymbol{R}_n$ vector is evaluated in the step, and gives us a stability criterion

$$\gamma h < 4. \tag{7.46}$$

A similar calculation and linear solve is needed to compute $\boldsymbol{\lambda}_{n+3/4}$ in order to maintain the hidden constraint for $\boldsymbol{p}_{n+3/4}$, whereas the Lagrange multiplier $\boldsymbol{\lambda}_{n+1/2}$ is responsible for ensuring that $\boldsymbol{g}(\boldsymbol{q}_{n+1})=\mathbf{0}$, and is typically found using an implicit solve.

There are many other ways we could choose to combine traditional molecular dynamics discretization schemes with constrained solvers such as variants of the SHAKE algorithm (see, for example [81, 383, 396]). The following section presents an alternative based upon the work on splitting methods in this chapter so far.

7.12.1 A "Geodesic Integrator"

We may consider alternative splittings of the dynamics in order to build a library of available schemes, just as we did in the unconstrained case in Sect. 7.3.1. Solving the constrained OU part exactly would allieviate the condition (7.46), while we may hope to reduce discretization error in the spirit of the success of the ⟦BAOAB⟧ scheme in Sect. 7.9.3. The natural splitting template to consider would be to again split into potential and kinetic energies, with the constrained Ornstein-Uhlenbeck piece:

$$\begin{bmatrix}\mathrm{d}\boldsymbol{q}\\ \mathrm{d}\boldsymbol{p}\\ \mathbf{0}\end{bmatrix}=\underbrace{\begin{bmatrix}\boldsymbol{M}^{-1}\boldsymbol{p}\,\mathrm{d}t\\ -\boldsymbol{g}'(\boldsymbol{q})^T\boldsymbol{\lambda}\,\mathrm{d}t\\ \boldsymbol{g}(\boldsymbol{q})\end{bmatrix}}_{\mathrm{A}_c}+\underbrace{\begin{bmatrix}\mathbf{0}\\ -\nabla U(\boldsymbol{q})\mathrm{d}t-\boldsymbol{g}'(\boldsymbol{q})^T\boldsymbol{\lambda}\,\mathrm{d}t\\ \boldsymbol{g}(\boldsymbol{q})\end{bmatrix}}_{\mathrm{B}_c}$$
$$+\underbrace{\begin{bmatrix}\mathbf{0}\\ -\gamma\boldsymbol{p}\mathrm{d}t+\sqrt{2\gamma k_{\mathrm{B}}\mathrm{T}\boldsymbol{M}}\mathrm{d}\mathbf{W}-\boldsymbol{g}'(\boldsymbol{q})^T\boldsymbol{\lambda}\,\mathrm{d}t\\ \boldsymbol{g}(\boldsymbol{q})\end{bmatrix}}_{\mathrm{O}_c}.$$

Observe that the holonomic constraint (as well as the hidden constraint) is maintained in each part, changing the usual A piece into a *geodesic drift* along the

manifold defined by $\boldsymbol{g}(\boldsymbol{q}) = 0$. We will denote schemes using this splitting in a similar fashion to the unconstrained case, but precede the schemes with "c-" to label the constrained form (for example c-⟦BAOAB⟧ rather than the more cumbersome notation $[\![\mathrm{B_c A_c O_c A_c B_c}]\!]$). Such a constrained scheme would be:

$$\textbf{c-}[\![\mathrm{BAOAB}]\!]$$

$$\text{Solve} \begin{cases} \boldsymbol{p}_{n+1/5} = \boldsymbol{p}_n - \frac{h}{2}\nabla U(\boldsymbol{q}_n) - \boldsymbol{g}'(\boldsymbol{q}_n)^T \boldsymbol{\lambda}_{n+1/3} \\ \text{Such that } \boldsymbol{g}'(\boldsymbol{q}_n)\boldsymbol{M}^{-1}\boldsymbol{p}_{n+1/5} = \boldsymbol{0} \end{cases}$$

Solve the geodesic flow A_c for $h/2$, and map $[\boldsymbol{q}_n, \boldsymbol{p}_{n+1/5}] \to [\boldsymbol{q}_{n+1/2}, \boldsymbol{p}_{n+2/5}]$

$$\text{Solve} \begin{cases} \boldsymbol{p}_{n+3/5} = e^{-\gamma h}\boldsymbol{p}_{n+2/5} + \sqrt{(1-e^{-2\gamma h})k_{\mathrm{B}}\mathrm{T}\boldsymbol{M}}\boldsymbol{R}_n - \boldsymbol{g}'(\boldsymbol{q}_{n+1/2})^T \boldsymbol{\lambda}_{n+2/3} \\ \text{Such that } \boldsymbol{g}'(\boldsymbol{q}_{n+1/2})\boldsymbol{M}^{-1}\boldsymbol{p}_{n+3/5} = \boldsymbol{0} \end{cases}$$

Solve the geodesic flow A_c for $h/2$, and map $[\boldsymbol{q}_{n+1/2}, \boldsymbol{p}_{n+3/5}] \to [\boldsymbol{q}_{n+1}, \boldsymbol{p}_{n+4/5}]$

$$\text{Solve} \begin{cases} \boldsymbol{p}_{n+1} = \boldsymbol{p}_{n+4/5} - \frac{h}{2}\nabla U(\boldsymbol{q}_{n+1}) - \boldsymbol{g}'(\boldsymbol{q}_{n+1})^T \boldsymbol{\lambda}_{n+1} \\ \text{Such that } \boldsymbol{g}'(\boldsymbol{q}_{n+1})\boldsymbol{M}^{-1}\boldsymbol{p}_{n+1} = \boldsymbol{0}, \end{cases}$$

where $\boldsymbol{R}_n$ is a vector of i.i.d $\mathcal{N}(0,1)$ random numbers and the Lagrange multipliers $\boldsymbol{\lambda}$, ensuring that the hidden constraint is satisfied, can be computed through a linear solve. We do not necessarily need to solve the geodesic flow exactly in practice, though for the purposes of analysis we shall assume we do. If we are considering a second order scheme, then solving the A_c piece to at least second order would be sufficient to preserve the overall order of the scheme, although a scheme may have potentially better accuracy and stability properties if a higher order solve is used.

In practice, we can solve the A_c piece using any constraint algorithm for constant energy dynamics (see Sect. 4.3.6) and subsituting zero force. A more accurate solution is obtained for the geodesic drift by either using a number of small steps to integrate over the total time, or composing steps in a certain order to obtain a higher order solution (see Sect. 3.3.3). Relative to the force calculation, solving the geodesic drift step (even to high accuracy) can, in typical molecular systems, be considered computationally cheap. Given these splitting pieces, one may write down constrained equivalents of any of the schemes described in Sect. 7.3.1.

We then ask how to analyze such schemes using the algebraic machinery developed throughout this chapter. As we solve each vector field on the constraint manifold, it is natural to consider the associated Lie derivatives on these manifolds and solve the corresponding Fokker-Planck equation for the discretization to find an invariant distribution, just as we did in the unconstrained case. This is a complicated programme and we do not develop this in detail here.

To compare the performance of the various schemes we perform a numerical experiment to quantify the discretization error caused by the schemes. We implement the c-⟦BAOAB⟧ and c-⟦OBABO⟧ schemes into the NAMD-lite [170, 302] package and consider 233 TIP3P water molecules in a periodic box with dimensions $20 \times 20 \times 20$ Angstroms at 300 K. We constrain the angles and bond lengths of each molecule so that the only forces felt are due to the van der Waals and electronic interations, parameterized by the CHARMM22 forcefield [246] with an 8-Angstrom cutoff. The system was simulated for a fixed total time of 3ns, after a